

Package ‘HPLNC’

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bathy.basemap

Create a Bathymetric and Topographic Basemap

Description

(Add your description here...)

Usage

```
bathy.basemap(
  boundaries = list(0, -100, 75, 20),
  vertical.tiles = 5,
  horizontal.tiles = 5,
  checkerboard = T,
  bathy.clr = T,
  land = T,
```

```

    crs = 4326,
    resolution = 5,
    z = 3,
    cache = T
)

```

Arguments

boundaries	A list of boundaries (min.lat, min.long, max.lat, max.long). Default values are suitable for North Atlantic.
vertical.tiles	Number of vertical checkerboard tiles. Will be ignored if checkerboard if False.
horizontal.tiles	Number of horizontal checkerboard tiles. Will be ignored if checkerboard if False.
checkerboard	Logical. Add checkerboard scales?
bathy.clr	Logical. Add bathymetry colors?
land	Logical. Add land elevation colors?
crs	The coordinate reference system.
resolution	Resolution for getNOAA.bathy.
z	Zoom level for elevatr::get_elev_raster.

clean_label	<i>Clean Sequence Labels in Alignment</i>
-------------	---

Description

This only applies to DZMB collection where the sequences are managed using field ID. The gene will be determined by the file name (whether it's COI, 18S, or 28S) and a gene suffix will be added to the voucher (e.g. VPS101_COI, ZHH003_18S).

Usage

```
clean_label(fasta_path)
```

Arguments

fasta_path	Path to FASTA file
------------	--------------------

Value

A new FASTA file will be written in the same directory. The path to the new file will be returned.

clearRCache	<i>Clear temporary cache file of R</i>
-------------	--

Description

Clear temporary cache file of R

Usage

```
clearRCache()
```

create_log	<i>Initialize daily log</i>
------------	-----------------------------

Description

Initialize daily log

Usage

```
create_log()
```

create_summary_sp	<i>Create Species Summary Report in Image and PDF</i>
-------------------	---

Description

Create Species Summary Report in Image and PDF

Usage

```
create_summary_sp()
```

date2assets	<i>Create assets folder for the day</i>
-------------	---

Description

Create assets folder for the day

Usage

```
date2assets(date)
```

DBchange_sign	<i>Generate Signature for Database Change</i>
---------------	---

Description

Generate Signature for Database Change

Usage

```
DBchange_sign(  
  table,  
  type,  
  msg,  
  request = request.id,  
  con = dbConnect(RSQLite::SQLite(), "data/database/HPLNCdb.sqlite"),  
  time = format(Sys.time(), "%Y-%m-%d %H:%M:%S %Z"),  
  signature.table = "signature"  
)
```

Arguments

table	Table name subjected to change
type	Change type
request	Request ID
con	Connection to database as produced by DBI::dbConnect()
time	Time of change. Default to current time.
signature.table	Table name to write signature.

DBexecute	<i>Execute YAML Change Request</i>
-----------	------------------------------------

Description

Execute YAML Change Request

Usage

```
DBexecute(request.filename)
```

DBlastRequest	<i>Return Signature of the Last Request</i>
---------------	---

Description

Return Signature of the Last Request

Usage

```
DBlastRequest()
```

DBmanual_pull	<i>Pull Raw Table</i>
---------------	-----------------------

Description

Pull Raw Table

Usage

DBmanual_pull(table)

Arguments

table Table name.

DBmanual_write	<i>Write Table</i>
----------------	--------------------

Description

Write Table

Usage

DBmanual_write(tbl, table, msg, request.id)

Arguments

tbl Table object to write.
table Table name.
msg Change message.
request.id Request ID associated to this change.

DBpullTable	<i>Pull Table</i>
-------------	-------------------

Description

Pull Table

Usage

DBpullTable(table, cleaned = T, formatting = T, format = NULL)

Arguments

table	Table name. Type "show_table" to display all available table names in database.
cleaned	Clean table, rows marked in del column will be dropped.
formatting	Format table. Utilize table-specific formatting function to format the table if available. Formatting functions are always named under the rule format_ + table name.
format	Argument to pass over to formatting functions. If available different formatting could be chosen.

DBsnapshot	<i>Generate Snapshot from All Tables</i>
------------	--

Description

Generate Snapshot from All Tables

Usage

DBsnapshot()

dec_format	<i>Format DD Coordinates with Letters.</i>
------------	--

Description

Format decimal degree coordinates notated with letters to clean numerical coordinates. West and south with minus.

Usage

dec_format(dec)

Arguments

dec	DD coordinates. This is a vectorized function, which means you can either pass one character that is latitude or longitude coordinates, or you can pass a vector of those. But not both latitude and longitude at the same time, it only handles one vector at a time.
-----	--

Author(s)

Zhehao Hu

dms_to_decimal	<i>Convert DMS or DDM Coordinates to DD Coordinates.</i>
----------------	--

Description

Also handles various notations and direction letter just as they are written in the cruise reports, which is never ready to use.

Usage

```
dms_to_decimal(dms)
```

Arguments

dms	DMS or DDM coordinates. This is a vectorized function, which means you can either pass one character that is latitude or longitude coordinates, or you can pass a vector of those. But not both latitude and longitude at the same time, it only handles one vector at a time.
-----	--

Value

Converted decimal degree coordinates, ready to use.

Author(s)

Zhehao Hu

extract_INSD_metadata	<i>Extract INSD Metadata from xml File Downloaded From NCBI Genebank</i>
-----------------------	--

Description

Extract INSD Metadata from xml File Downloaded From NCBI Genebank

Usage

```
extract_INSD_metadata(xml)
```

Arguments

xml	File path to xml file
-----	-----------------------

Value

A data.frame of flattened tabular metadata.

fetch.land.elv	<i>Fetch Land Elevation Data</i>
----------------	----------------------------------

Description

Fetch Land Elevation Data

Usage

```
fetch.land.elv(boundaries, crs = 4326, z = 3, cache = T)
```

Arguments

boundaries	A list of boundaries (min.lat, min.long, max.lat, max.long).
crs	Coordinate reference system.
z	Zoom level for elevation.
cache	Logical. Default to TRUE, the function will generate cache for the exact same boundary for future reuse.

fetch.NOAA.bathy	<i>Fetch bathymetry map from NOAA.</i>
------------------	--

Description

Fetch bathymetry map from NOAA.

Usage

```
fetch.NOAA.bathy(boundaries, resolution = 5, cache = T)
```

Arguments

cache	Logical. Default to TRUE, the function will generate cache for the exact same boundary for future reuse.
-------	--

format_metadata.Sequence.NCBI	<i>Format NCBI Sequence Metadata</i>
-------------------------------	--------------------------------------

Description

Format NCBI Sequence Metadata

Usage

```
format_metadata.Sequence.NCBI(tbl, formatting = T, format = 1)
```

```
format_metadata.Specimen.Haploniscidae
```

Format Specimen Metadata

Description

Format Specimen Metadata

Usage

```
format_metadata.Specimen.Haploniscidae(tbl, formatting = T)
```

```
format_metadata.Station
```

Format Station Metadata

Description

Format Station Metadata

Usage

```
format_metadata.Station(tbl, formatting = T)
```

```
format_morphocheck
```

Format morphocheck result

Description

Format morphocheck result

Usage

```
format_morphocheck(tbl)
```

gblocks

*Run Gblocks to Trim the Alignment***Description**

This is a wrapper function to call Gblocks in CLI. For details about the program please visit: https://home.cc.umanitoba.ca/~psgendb/doc/Castresana/Gblocks_documentation.html

Usage

```
gblocks(
  fasta_path,
  gblocks_path = "/Users/hu_zhehao/Desktop/Biology-tools/Gblocks_0.91b/Gblocks",
  t = "d",
  b1 = "min",
  b2 = "min",
  b3 = 20,
  b4 = 2,
  b5 = "a",
  args = ""
)
```

Arguments

fasta_path	File path of the alignment FASTA file.
gblocks_path	File path of the Gblocks executable.
t	Type of sequence. Gblocks command line parameter, will be passed over as -t.
b1	Minimum number of sequence for a conserved position ($\geq 50\%+1$ of the number of sequences, default $50\%+1$ of the number of sequences). Gblocks command line parameter, will be passed over as -b1. Also accepts min to choose the lowest value, which maximize the selected number of positions (i.e. least aggressive in trimming).
b2	Minimum number of sequences for a flank position ($\geq b1$, default 85% of the number of sequences). Gblocks command line parameter, will be passed over as -b2. Also accepts min to choose the lowest value, which maximize the selected number of positions (i.e. least aggressive in trimming).
b3	Maximum number of contiguous nonconserved positions (gblocks default to 8, here default to 20). Bigger values increase the selected number of positions.
b4	Minimum length of a block after gap cleaning (gblocks default to 10, minimum value 2, here default to 2). Bigger values decrease the selected number of position.
b5	Allow gap positions. n for absolute no gap positions allowed in the final alignment. h for only positions where 50% or more of the sequens have a gap are treated as gap position and are eliminated. a for all gap positions can be selected, no positions are eliminated because of the gaps.
args	Other arguments that are passed directly to CLI command.

Value

Trimmed alignment will be written in new FASTA file under the same directory with suffix `_gbtrimmed.fasta`. The path to the new file will be returned.

<code>geom_checkerboard</code>	<i>Add checkerboard to a ggplot</i>
--------------------------------	-------------------------------------

Description

Add checkerboard to a ggplot

Usage

```
geom_checkerboard(
  min.lat,
  max.lat,
  lat.interval.count,
  min.long,
  max.long,
  long.interval.count,
  scale_only = F
)
```

Arguments

<code>scale_only</code>	Logical. If TRUE no checkerboard will be drawn, but the plot will be scaled (cropped or expanded) to the area according to giving boundaries.
-------------------------	---

Value

Updated ggplot objects.

<code>ID.inquiry</code>	<i>Query specimen info with DZMB2HH ID</i>
-------------------------	--

Description

Query specimen info with DZMB2HH ID

Usage

```
ID.inquiry(ID)
```

load_morphocheck	<i>Load latest morphocheck result</i>
------------------	---------------------------------------

Description

Load latest morphocheck result

Usage

```
load_morphocheck()
```

midwayTreeViz	<i>Snapshot the MrBayes Result During the Analysis</i>
---------------	--

Description

This function allows a visualization of a consensus tree with currently sampled tree without stopping the MrBayes process.

Usage

```
midwayTreeViz(
  assets_date,
  outgroup,
  subfolder = "",
  treeviz = T,
  xlim.factor = 1.6
)
```

Arguments

assets_date	Date of the asset folder where the MrBayes analysis process is located in. This is a project-specific data structure adaptation. It just simplifies the location of the target files.
outgroup	Sequence labels of the outgroup sequences. It will use automatic gap detection to find the sequences so it allows a certain degree of ambiguity in the name.
subfolder	Subfolder where the MrBayes process is located under the asset folder of the date. This is a project-specific data structure adaptation. It just simplifies the location of the target files.
treeviz	logical. Whether to visualize the tree. Default to TRUE.
xlim.factor	Factor to expand the x-axis of the tree. Increase this value if long labels are omitted because they exceed the plotting area.

Value

A ggplot of visualized tree if treeviz is set to TRUE. A tidytree "treedata" class object if treeviz is set to FALSE.

nexus2fasta

Convert NEXUS alignment to FASTA alignment

Description

Convert NEXUS alignment to FASTA alignment

Usage

```
nexus2fasta(
  nexus_path,
  fasta_path = NULL,
  safe_name = F,
  remove_outgroup = NULL
)
```

Arguments

nexus_path	Path to NEXUS file.
fasta_path	Path to FASTA file.
safe_name	logical. Whether to modify the name to remove "_".
remove_outgroup	Vector of sequence labels to remove. Default to NULL without removing any sequence. This happens before modifying sequence labels if safe_name is set to TRUE.

Value

The FASTA file will be written in the same directory. The path will be returned.

render_species_summary

Wrapper for Quarto Render Species Summary

Description

Wrapper for Quarto Render Species Summary

Usage

```
render_species_summary()
```

summary_sp	<i>Generate Species Summary</i>
------------	---------------------------------

Description

Generate Species Summary

Usage

```
summary_sp(df, metadata.station, metadata.specimen, species)
```

update_INSD_metadata	<i>Wrapper Function to Read All XML File and Update Information to Databse.</i>
----------------------	---

Description

Wrapper Function to Read All XML File and Update Information to Databse.

Usage

```
update_INSD_metadata(XML_folder = "data/sequence/INSD", regenerate = F)
```

Arguments

XML_folder	Path to folder containing xml files. All xml files in the folder will be loaded.
regenerate	logical. Whether to overwrite the table in database (i.e. regenerating the table). Default to FALSE, which will only add metadata that has not been added in the table.

Value

Nothing.

update_sequence_map	<i>Create a Summary Table Mapping Sequence Label/Accession Number with Vouchered Animals.</i>
---------------------	---

Description

This will summarize the sequence labels in the way that the sequences originated from the same vouchered animals could be linked together, which is essential for concatenate multilocus alignment.

Usage

```
update_sequence_map(regenerate = F)
```

Arguments

regenerate	logical. Whether to overwrite existing table in the database. Otherwise only the rows with vouchered animal not yet in the database will be added.
------------	--

Value

Nothing. A data.frame will be written into the database.

visualize_tree	<i>Visualize Consensus Tree Produced From MrBayes Analysis</i>
----------------	--

Description

Visualize Consensus Tree Produced From MrBayes Analysis

Usage

```
visualize_tree(assets_date, outgroup, subfolder = "", xlim.factor = 1.5)
```

Arguments

assets_date	Date of the asset folder where the MrBayes analysis process is located in. This is a project-specific data structure adaptation. It just simplifies the location of the target files.
outgroup	Sequence labels of the outgroup sequences. It will use automatic gap detection to find the sequences so it allows a certain degree of ambiguity in the name.
subfolder	Subfolder where the MrBayes process is located under the asset folder of the date. This is a project-specific data structure adaptation. It just simplifies the location of the target files.
xlim.factor	Factor to expand the x-axis of the tree. Increase this value if long labels are omitted because they exceed the plotting area.

Value

A ggplot of visualized tree.

voucher.inquiry	<i>Query specimen info with voucher</i>
-----------------	---

Description

Query specimen info with voucher

Usage

```
voucher.inquiry(ID)
```


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